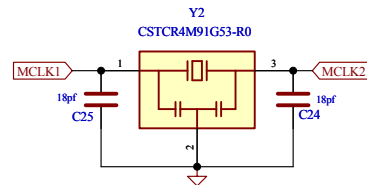
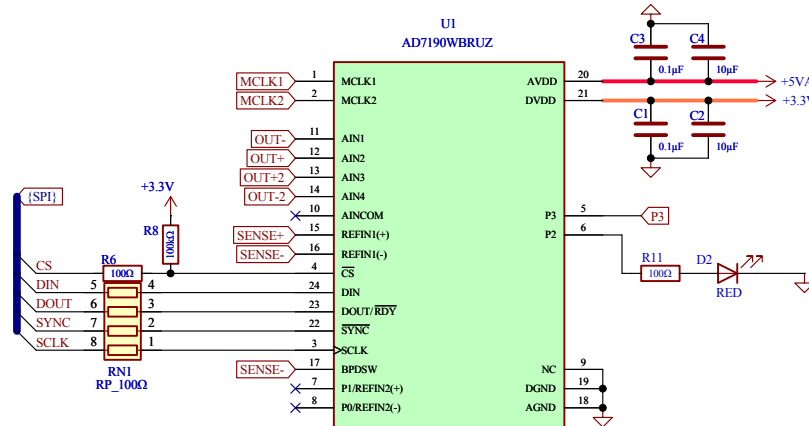


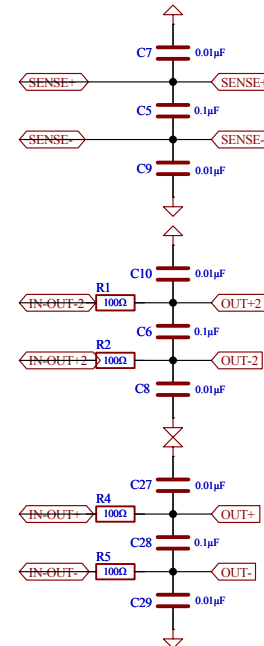
# [4] AD7190 - Weight Sensor



## DESIGN NOTE:

Ref Condo for quartz  
Murata GRM1885C1H220JA01D  
(22 pF, 50 V, C0G, 0603, ±5%)

## ADC FOR weighing sensor



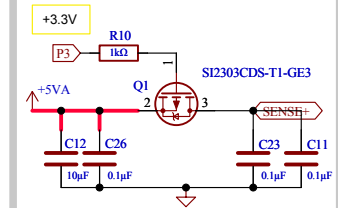
Calcul de filtre :  $f_c = 1 / (2\pi \times R \times C) = X \text{ Hz}$

## DESIGN NOTE:

Ref Condo for Filtre  
Murata GRM1885C1H103JA01D (10 nF, 50 V, C0G, 0603, ±5%)  
KEMET C0603R104K5RACTU (100 nF, 50 V, X7R, 0603, ±10%)

Ref Resistor for Filtre  
TNPW0603100RBEEA (100 Ω, 0.1 W, thin film, 0603, ±0.1%, 25 ppm/°C)

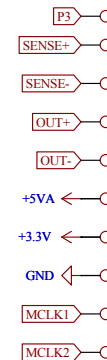
## FILTRE



## DESIGN NOTE:

IC1 is a P-MOS high-side switch used to enable/disable the load cell excitation. must disabled during calibration

## DRIVE GATE



## TEST POINT

## About

The AD7190 is a low noise, complete analog front end for high precision measurement applications. It contains a low noise, 24-bit sigma-delta ( $\Sigma\Delta$ ) analog-to-digital converter (ADC). The on-chip low noise gain stage means that signals of small amplitude can be interfaced directly to the ADC. The device can be configured to have two differential inputs or four pseudo differential inputs. The on-chip channel sequencer allows several channels to be enabled, and the AD7190 sequentially converts on each enabled channel. This simplifies communication with the part. The on-chip 4.92 MHz clock can be used as the clock source to the ADC or, alternatively, an external clock or crystal can be used. The output data rate from the part can be varied from 4.7 Hz to 4.8 kHz. The device has two digital filter options. The choice of filter affects the rms noise/noise-free resolution at the programmed output data rate, the settling time, and the 50 Hz/60 Hz rejection.



## Comments:

AD7190  
CSTCR4M91G53-R0  
AD7190\_TEST\_BOARD CN

## Company:

OPEN THRUST LAB

## Board Name:

AD7190

## Sheet Title:

AD7190 - Weight Sensor

## Sheet Path:

/Project Architecture/AD7190\_Weight\_Sensor/

## File Name:

AD7190 - Weight Sensor.kicad\_sch

## Designer:

SIEBERT DIMITRY

## Reviewer:

## Variant:

OPEN THRUST LAB

## Project Name:

OPEN\_WEIGHT

## Date:

2026-09-29

## Size:

A4

## Revision:

+

## Sheet:

4 of 10

## Git Hash:

b3867f9